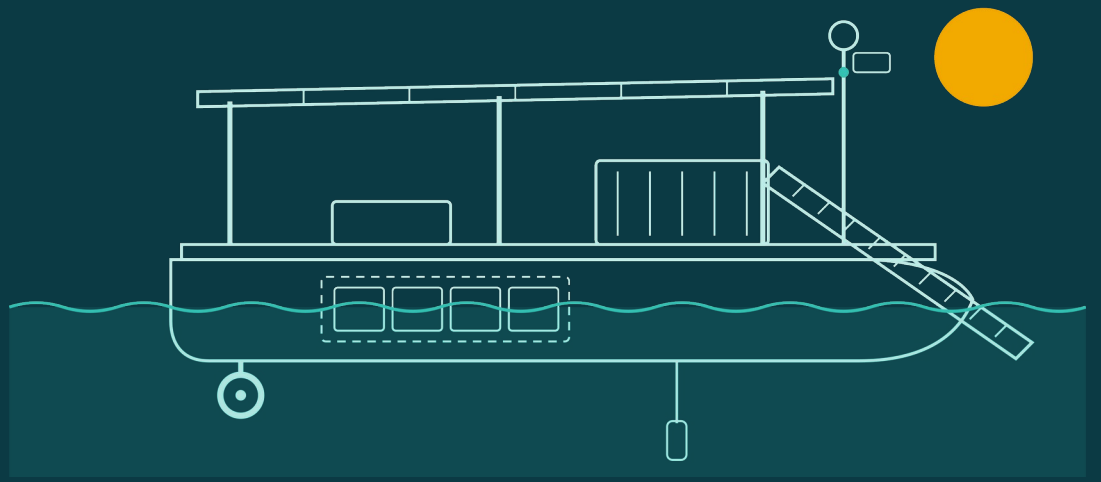
CONCEPT DESIGN BRIEF · NOVEMBER 2026

Project SAGAR

Solar boats and a Swiss-style interceptor plant to bring Hussain Sagar back to life



Target: Hussain Sagar, Hyderabad

SAGAR = Solar Autonomous Grid for Aquatic Restoration
Prepared for product design engineering review

EXECUTIVE SUMMARY

A sewage-fed lake that can be fixed

Hussain Sagar gets at least 40 million litres of untreated water a day. SAGAR stops it at the inlet and cleans the surface with solar boats.

The problem

- ≥ 40 million litres of untreated water enter every day
- Oxygen averages ~1.9 mg/L; fish need ≥ 4
- Organic load (BOD) is 8–25× the bathing limit
- Sept 2026 immersion: still not recovered

The solution

- SAGAR-1: a 100 MLD Swiss-style plant built across Kukatpally nala
- 12 solar boats skim debris and algae and polish hotspots
- Sondes and buoys feed a live public dashboard

The ask

- Engineer one solar boat platform with two payloads
- Build 3 prototypes in 9 months
- Field-test at the Ganesh immersion, Sept 2027

01

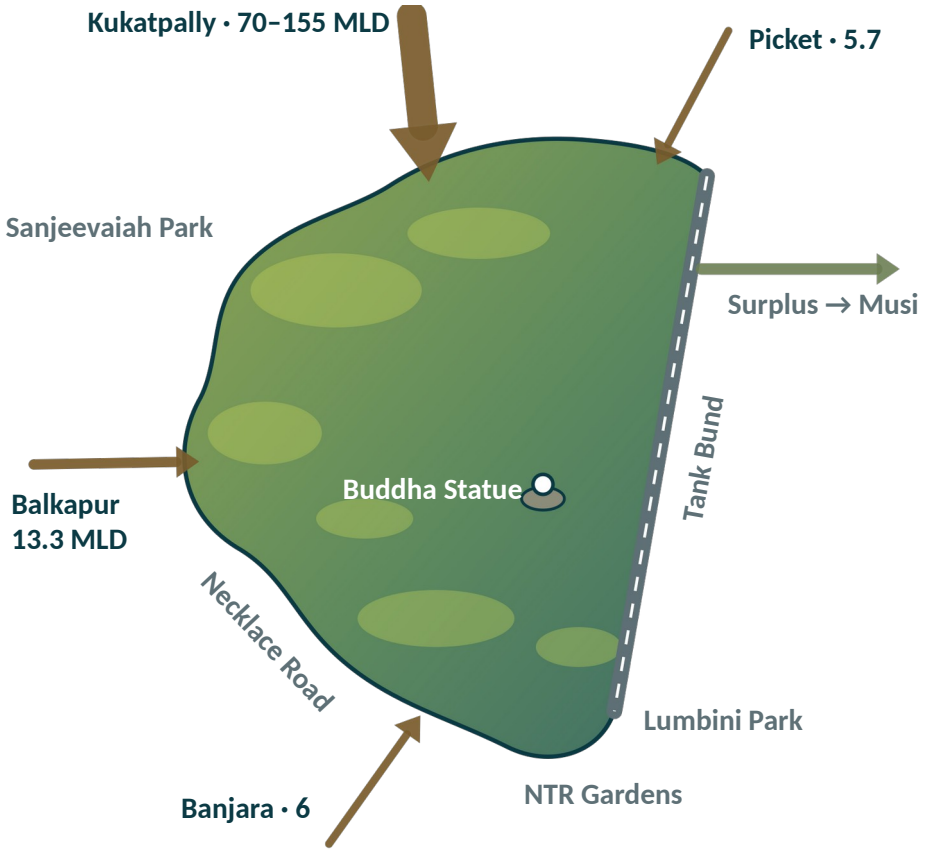
The problem

What the latest verified data says about Hussain Sagar



Hussain Sagar at a glance

A 460-year-old reservoir in the heart of Hyderabad, now fed mainly by four sewage drains.



1562
Built under Ibrahim Quli Qutb Shah

240 km²
Catchment draining into the lake

4.7–5.7 km²
Water-spread area (sources differ)

23.5–27.2 Mm³
Volume: 23.5 to 27.2 billion litres

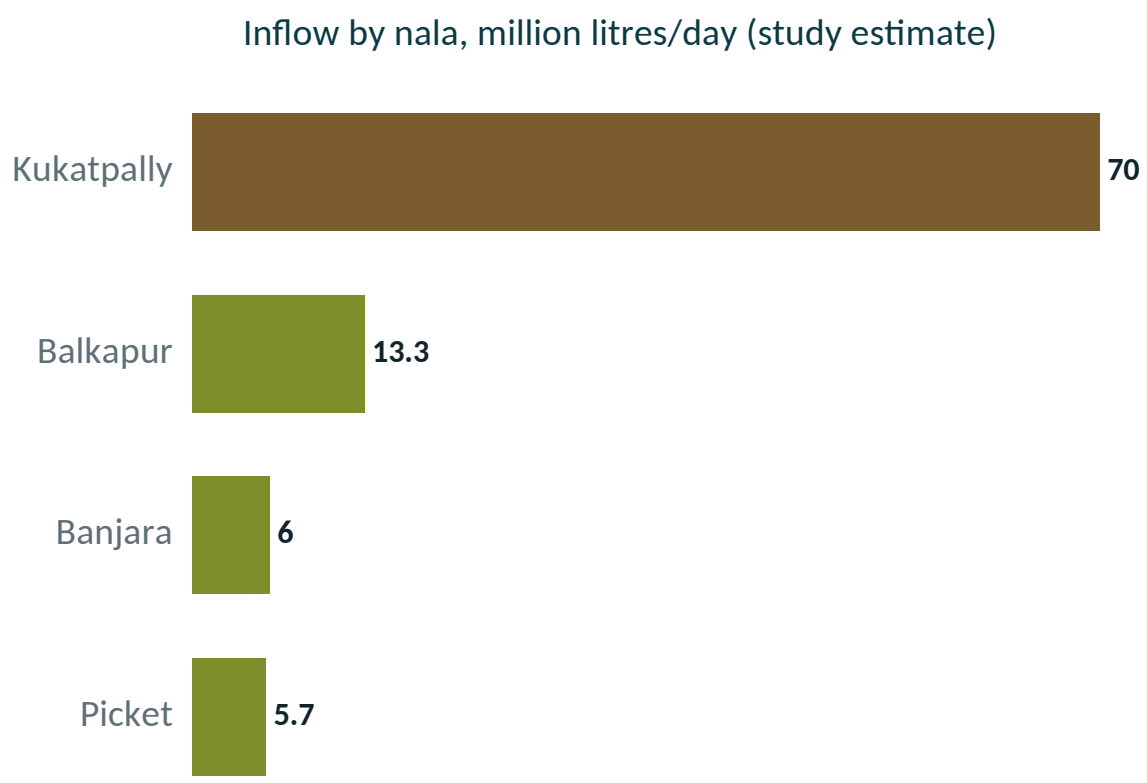
~9.8 m
Maximum depth (32 ft)

4 nalas
Feeder drains, all carrying sewage

Schematic map, not to scale; arrow width ∝ inflow (study estimate). Sources: HMDA presentation; Telangana Today; Wikipedia; neliti review of Hussain Sagar pollution.

Four drains carry sewage in

Kukatpally nala dominates: 70–155 MLD of mixed domestic sewage and industrial effluent.



Kukatpally nala

Mixed sewage and industrial effluent from the north. 70 MLD in a study; 155 MLD reported (Telangana Today, 2020).

Picket, Banjara, Balkapur

Mainly domestic sewage. About 25 MLD combined on the study basis.

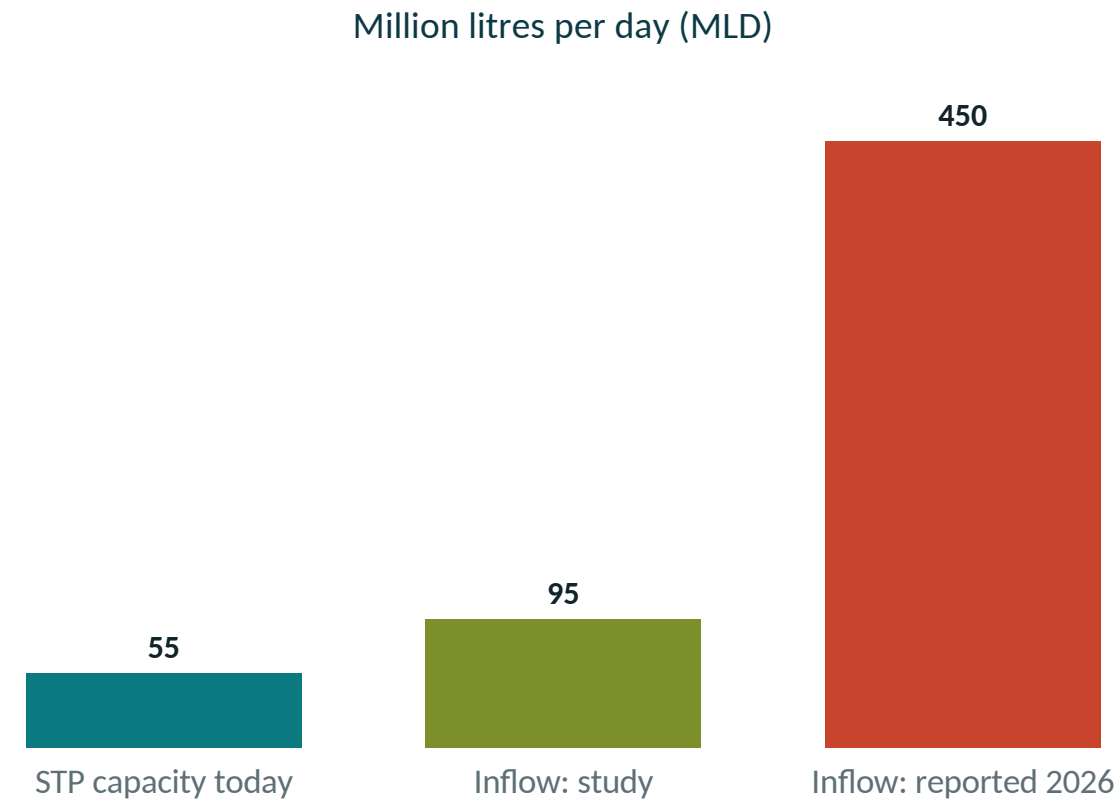
Reported total: ~450 MLD

The Federal, Feb 2026. Method not published. Phase 0 meters all four nalas to settle it.

Sources: Review of Hussain Sagar Lake Pollution (neliti); Telangana Today, 15 Dec 2020; The Federal, 18 Feb 2026; HCAIP.

Treatment covers about half

Even on the lowest estimate, at least 40 MLD enters untreated: about 16 Olympic pools a day.



≥ 40 MLD

enters untreated every day on the lowest estimate (95 in, 55 treated)

~9.5 months

for the nalas to deliver a volume equal to the whole lake (~2 months at 450 MLD)

0

verified, current flow measurements found for the four nalas. Phase 0 fixes this

STP capacity 5 + 20 + 30 MLD (Telangana Today, Nov 2022; a planned upgrade to 110 MLD is unconfirmed). Lake volume 27.18 Mm^3 (HMDA). Derived: 27.18 Mm^3 ÷ 95,000 m^3/day ≈ 286 days.

Fails even the fish standard

Typical oxygen is below what fish need, and the organic load is 8–25× the bathing limit.

Parameter	Measured in Hussain Sagar	Limit	Verdict
Dissolved oxygen	1.28–3.0 mg/L (2016–20); nil at 3 immersion sites (Sept 2025)	≥ 4 fish · ≥ 5 bathing	Fails
BOD	24–76 mg/L typical (pre-2020); 94 mg/L peak (Sept 2025)	≤ 3 mg/L bathing	8–25× over
Total coliform	1,600 MPN/100 mL at 8 of 9 points (2023)	≤ 500 bathing	3.2× over
COD	Peaks of 164 mg/L (2026) and 280 mg/L (2025)	≤ 50 mg/L (NGT discharge norm)	3–6× over
Turbidity	200 NTU peak, Necklace Road (Sept 2026)	No CPCB lake limit	Very high

1.9

mg/L

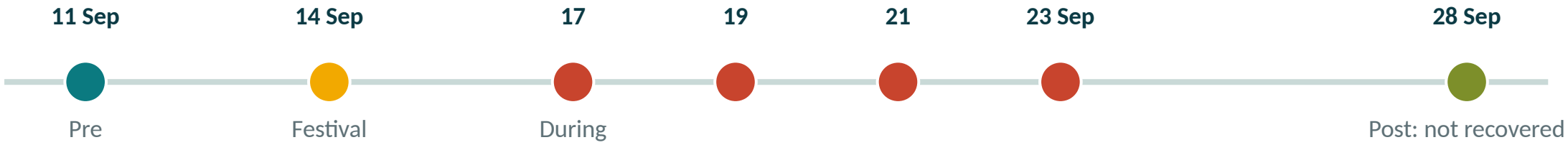
typical dissolved oxygen
2016–20 average

- **Fish need ≥ 4**
- **Bathing needs ≥ 5**
- **During immersion: 0**

Sources: IJARIIT (2016–20); TSPCB via Siasat (2020); TGPCB via Telangana Today (Oct 2025), Deccan Chronicle (Oct 2026, Jun 2023). Limits: CPCB criteria; NGT 2019 norms.

Immersion 2026: not recovered

TGPCB sampled six sites before, during and after the festival on 14 September 2026. Every key indicator rose.



200 NTU

turbidity peak

Necklace Road

164 mg/L

COD peak

Necklace Road

99 mg/L

suspended solids peak

Platform 1, NTR Park

6 of 6

sites saw turbidity, TSS, COD and BOD rise

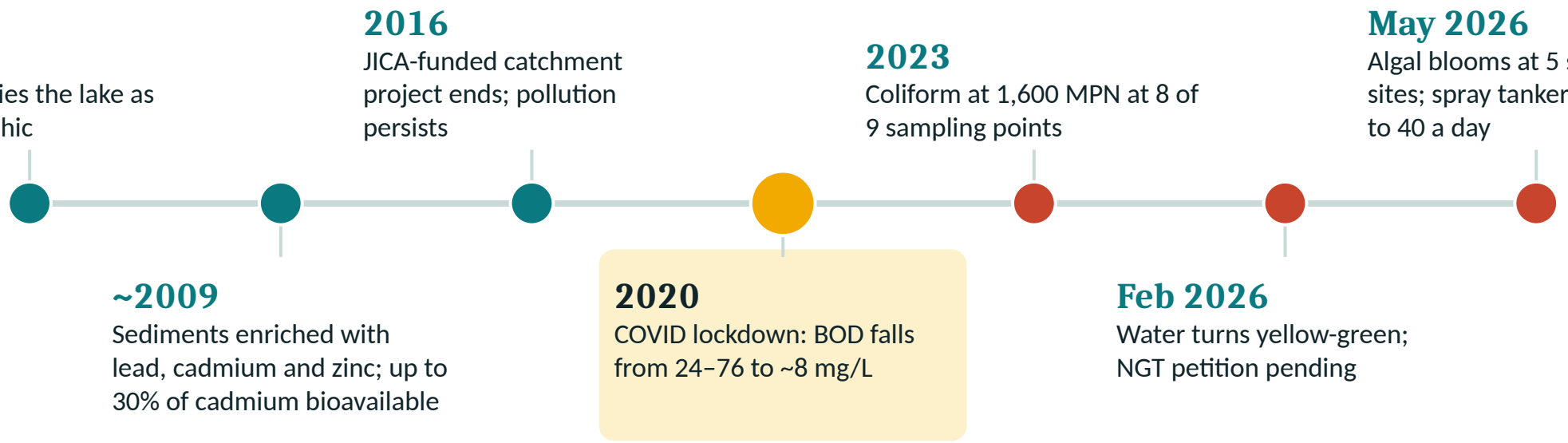
all monitoring points

For comparison, Sept 2025: dissolved oxygen fell to nil at 3 sites; BOD hit 94 mg/L and COD 280 mg/L at Platform 1.

Sources: TGPCB analysis reported by Deccan Chronicle and Telangana Today, 30 Oct 2026; TGPCB via Telangana Today, 5 Oct 2025.

Two decades of decline, one clue

When inflow stopped in the 2020 lockdown, BOD fell from 24–76 mg/L to about 8. Source control works.



Sources: Wikipedia (2008 study); Springer Env. Monit. Assess. (sediments); HCAIP; TSPCB via Siasat (2020); Deccan Chronicle (2023); The Federal (Feb 2026); Hyderabad Mail (18 May 2026).

Four design principles from the data

The volume problem sits at the inlets; the visible problem floats on the surface.

1 Stop it at the source

Build treatment across the biggest drain so dirty water never reaches the lake.

Evidence: ≥ 40 MLD untreated; lockdown BOD drop

2 Clean the surface with boats

The lake is nearly still, so fixed barriers won't work. Use mobile solar boats.

Evidence: Interceptor judged unsuitable for low-flow water

3 Put oxygen back at hotspots

Immersion bays and algae zones are where oxygen crashes first.

Evidence: DO nil at 3 sites, Sept 2025

4 Measure everything, continuously

Every boat carries a sonde; buoys and flow meters fill the gaps.

Evidence: no verified current nala flow data

02

The inspiration

What Switzerland and Sweden do differently



INSPIRATION

Learning from Switzerland and Sweden

Both treat water hard at the point where it enters the river or sea, instead of cleaning up afterwards.

Switzerland: 4th stage

- Law in force since 1 Jan 2016
- ~100–140 plants add a stage over 20 years
- Target: ≥ 80% micropollutant removal with ozone or activated carbon
- Funded by ~CHF 9 per person per year

WHAT WE TAKE
Ozone + carbon stage in SAGAR-1; carbon cartridge on the O-1 boat

Sweden: Henriksdal

- Stockholm's plant is the world's largest underground plant, built in rock
- Becoming the world's largest membrane bioreactor (MBR)
- > 500,000 m^3/day for 1.6 M people by 2040

WHAT WE TAKE
Compact MBR core; covered plant with a public park on top

Solar river clean-up

- Ocean Cleanup Interceptor: 100% solar, rated 50 t/day
- WasteShark drone: ~500 kg/day, ~8 h per charge
- Fixed barriers need current: Jakarta called the Interceptor unsuitable for low-flow rivers

WHAT WE TAKE
Fixed interception at the nala; mobile solar boats on the still lake

Sources: admin.ch/Eawag (2015); McArdell, Eawag (2019); ICPR; Nordic Investment Bank (2016); GE Reports; Eco-Business (2020); RanMarine.

03

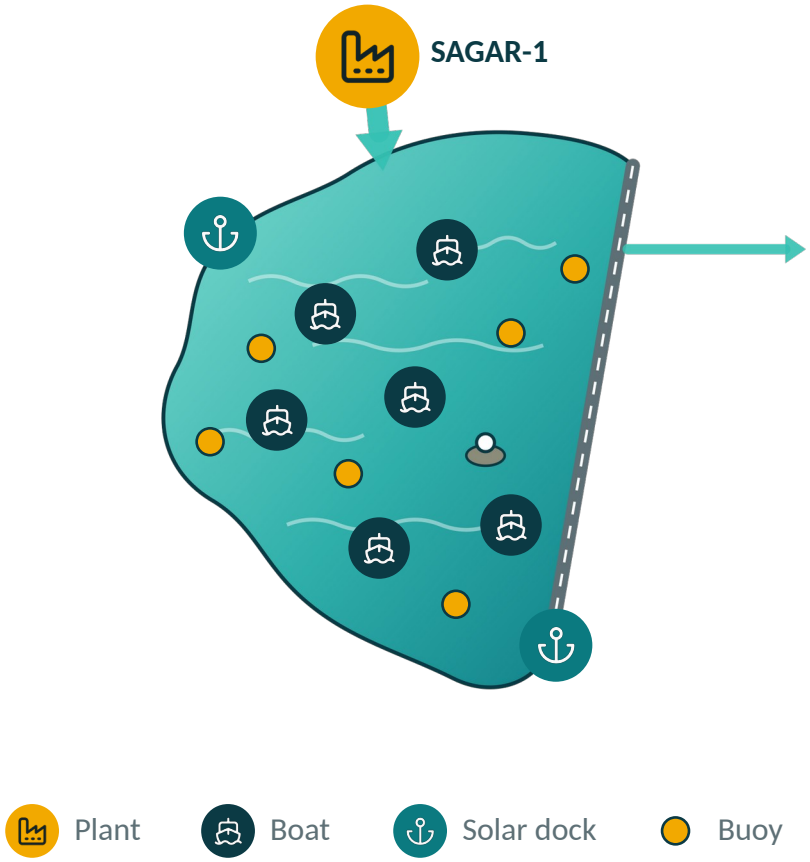
The solution

Project SAGAR: one plant, twelve solar boats, live data



One system, three layers

The plant stops the volume, the boats clean the surface, and sensors steer both.



STOP SAGAR-1 interceptor plant

100 MLD across Kukatpally nala: MBR, ozone + carbon, UV. Clean, oxygenated water flows into the lake.

CLEAN Solar boat fleet

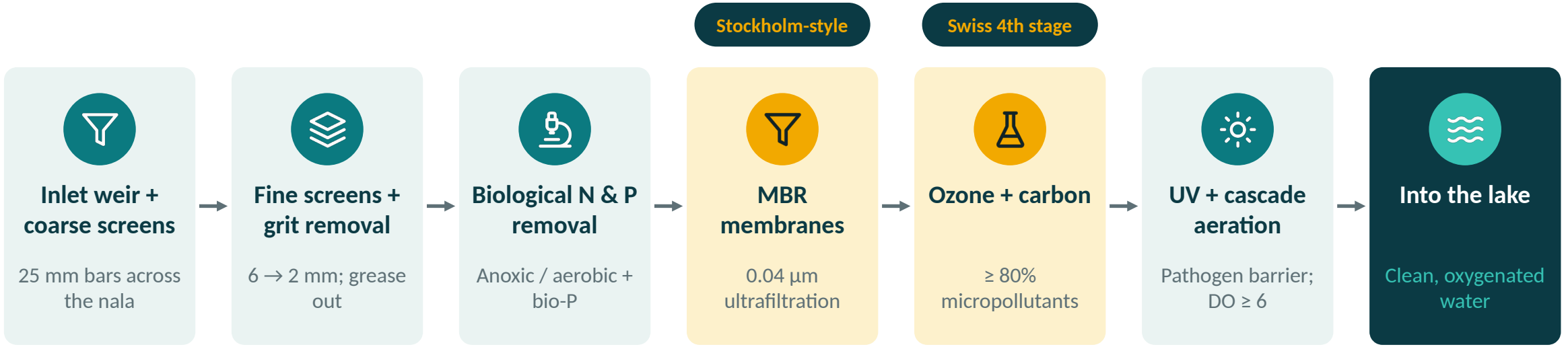
8 S-1 skimmers + 4 O-1 polishers from 2 solar docks. Debris, algae scum and hotspot oxygen.

SENSE Sondes, buoys, dashboard

DO, pH, conductivity, turbidity and chlorophyll on every boat, plus 6 buoys and nala flow meters, live online.

SAGAR-1: the plant across the nala

Every litre of dry-weather flow in Kukatpally nala passes five barriers before it reaches the lake.



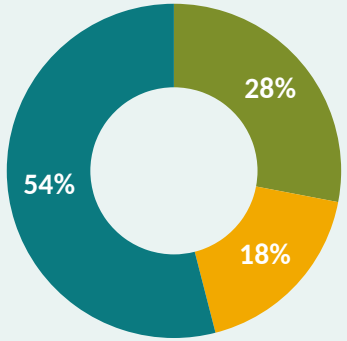
SIDE STREAMS

- Storm surplus overflows the weir after screening; first-flush tank holds the dirtiest runoff
- Sludge → digester → biogas power (~25–30% of plant energy)
- Phosphorus recovered as struvite fertiliser
- Plastics and screenings sent to recycling

Plant specification at a glance

Sized to cover the full reported range of Kukatpally flow, with modular trains to grow.

Parameter	NGT 2019 norm	SAGAR-1 target
BOD	≤ 10 mg/L	**≤ 5 mg/L**
COD	≤ 50 mg/L	**≤ 30 mg/L**
Suspended solids	≤ 20 mg/L	**≤ 2 mg/L**
Total nitrogen	≤ 10 mg/L	**≤ 8 mg/L**
Total phosphorus	≤ 1 mg/L	**≤ 0.3 mg/L**
Faecal coliform	< 230 MPN (desirable < 100)	**< 10 MPN/100 mL**
Micropollutants	Not regulated	**≥ 80% removed**



Energy: ~90 MWh/day

- Biogas CHP 25–30%
- Solar ≤ 4 MWp, ~18%
- Green-tariff grid, rest

100 MLD
average (150 peak), as 4 × 25 MLD trains

3.5–4 ha
covered footprint, park on the roof

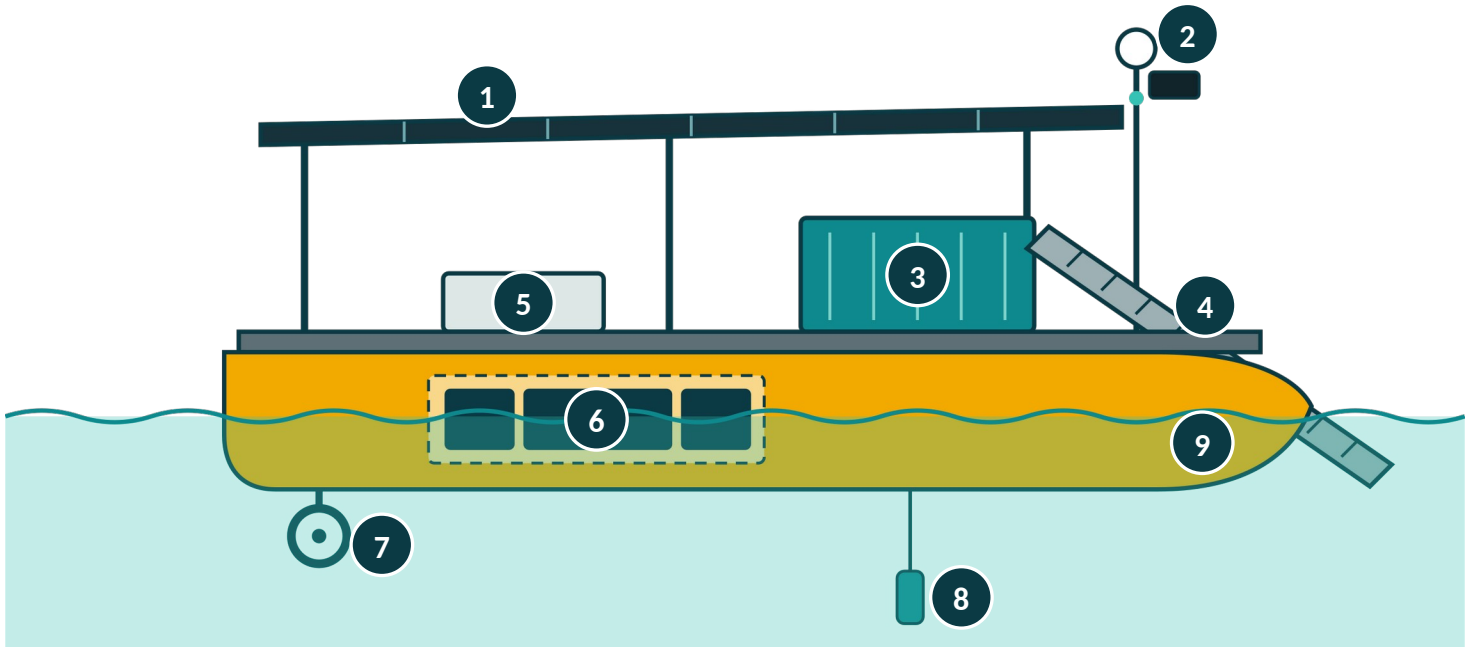
~0.9 kWh/m³
MBR + ozone + carbon + UV

₹450–550 cr
indicative capex (≈ ₹3.4 cr/MLD + premium)

Norms: NGT order (2019) for STPs in metro areas. Capex benchmark: HMWSSB 17-STP package, 376.5 MLD for ₹1,280.87 cr (Telangana Today, 2022). Energy and cost are first-pass estimates.

Meet the S-1 solar boat

A 4.2 m solar catamaran that skims, senses and runs all day on sunlight.

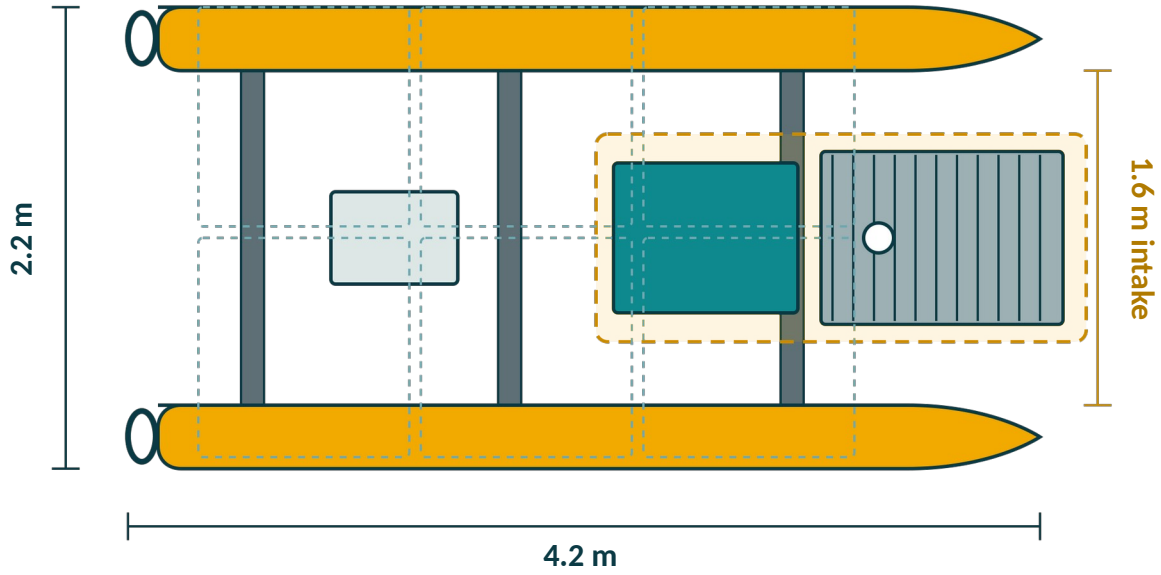


4.2 m long | **2.2 m beam** | **≤ 0.35 m draft** | **~380 kg** | **1–3 knots**

1. **Solar canopy** 1.62 kWp, 6 × 270 Wp
2. **Sensor mast** RTK-GNSS, stereo cameras, radar
3. **Debris bin** 250 L, ~120 kg per load
4. **Conveyor intake** 1.6 m wide, 5 mm mesh
5. **Electronics bay** compute, MPPT, LTE/LoRa, IP67
6. **Battery** 9.6 kWh LiFePO4, 4 hot-swap packs
7. **Rim-drive thrusters** 2 × 1 kW, no exposed blades
8. **Sonde** DO, pH, EC, turbidity, temp, chl-a
9. **Twin HDPE hulls** high-vis, foam-filled, unsinkable

One hull, two jobs

A common platform with swappable payloads: tool one boat, deploy it two ways.



Swappable payload bay (S-1 shown)

SHARED hull · solar · battery · propulsion · autonomy · sonde

S-1 Skimmer

An inclined mesh conveyor between the hulls lifts plastics, hyacinth, flowers, idol debris and algae scum into a 250 L bin.

1.6 m intake · ~400 kg/day · ~1.3 ha per shift

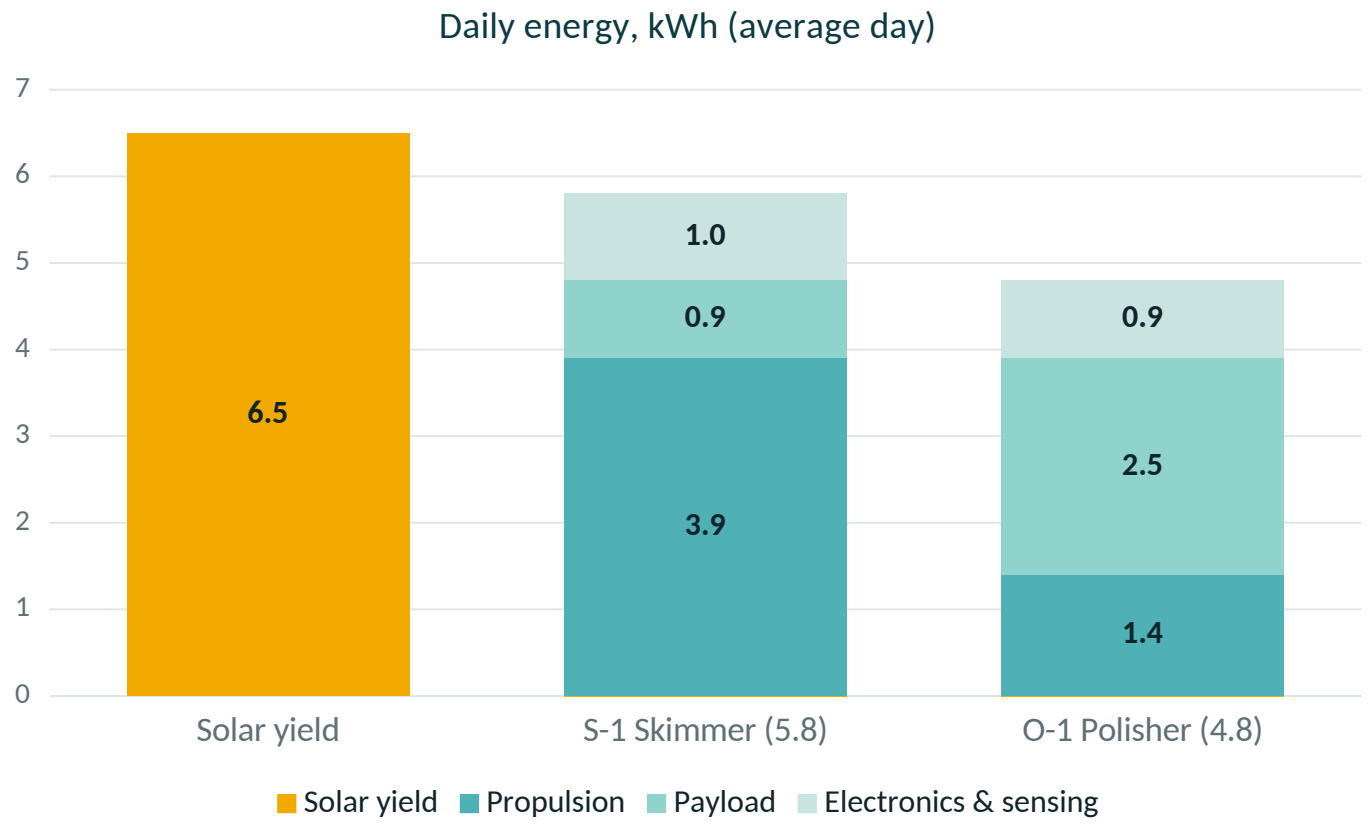
O-1 Polisher: the big plant, shrunk

Screen → Filter → Carbon → UV-C → O_2

10 m^3/h for immersion bays and algae hotspots

Runs all day on its own canopy

1.62 kWp of solar gives ~6.5 kWh on an average Hyderabad day, and both variants use less than that.



6.5 kWh/day

1.62 kWp × 5.0 peak-sun-hours × 0.80 performance ratio

1.3–1.6 days

of reserve with zero sun: 7.7 kWh usable from a 9.6 kWh pack set

Monsoon: ~4.9 kWh

at ~3.8 peak-sun-hours: shorten payload hours or swap packs at the solar dock

Load assumptions. S-1: skim 0.45 kW × 6 h, transit 0.8 kW × 1.5 h, conveyor 0.15 kW × 6 h. O-1: station-keeping 0.1 kW × 6 h, pump 0.30 + UV 0.12 kW × 6 h. Electronics 60 W active, 30 W standby.

12 boats, two docks, one dashboard

Cameras and sondes send boats to the debris and low-oxygen zones instead of mowing the whole lake.

Detect

Cameras, sondes and buoys flag debris and low oxygen

Dispatch

Fleet manager sends the nearest charged boat

Clean

Skim, polish or aerate, geofenced away from people

Dock

Swap packs, unload debris, upload data

3.2 t/day — debris capacity across 8 skimmers

~10 ha/day — swept by the skimmer fleet

~6 days — to revisit ~60 ha of hotspots: margins, nala mouths, immersion bays

2 shifts — in immersion season, running on swapped packs

Assumes 400 kg/day per skimmer (WasteShark benchmark: ~500 kg/day claimed) and 1.6 m × 1 knot sweep over a 4.5–6 h shift. Heavy idol remains still need shore cranes.

Why boats alone can't do it

The lake holds ~27 million m^3. Volume must be treated at the inlet; boats work the surface and hotspots.

27.2 million m^3

of water in the lake (HMDA, at spill level)

240 m^3/day

from 4 O-1 polishers combined: ~310 years to pass the lake's volume once

100,000 m^3/day

intercepted by SAGAR-1 at the inlet, so new pollution never arrives

With clean inflow, the lake flushes itself. 100 MLD of clean water replaces the lake's entire volume in about 9 months (27.18 Mm^3 ÷ 100,000 m^3/day ≈ 272 days).

04

Delivery

Requirements, open questions, cost, roadmap and targets



Design requirements

Hot, humid and UV-heavy, busy with tourist boats, and full of Plaster-of-Paris debris.

Environment

Air 10–46 °C, water 20–35 °C, UV index 11+, monsoon gusts ~60 km/h, pH 7–9.5

Materials

HDPE, 316 stainless and anodised aluminium only. No mild steel: conductivity ~1,700 µS/cm

Debris

Plastics, hyacinth, cloth, flowers, Plaster-of-Paris idol fragments, algae scum, biofouling

Safety

≤ 3 knots, geofence, e-stop on both sides, collision avoidance, bladeless rim-drives

Service

Payload swap < 15 min; 22 kg battery packs swap in < 2 min each, no tools

Manufacture

Rotomoulded hulls, off-the-shelf 48 V EV parts, ≥ 70% local content

Six open questions to settle

The trade-offs where the product designer's call matters most, each with our starting view.

1

Hull: rotomoulded HDPE or welded aluminium?

Starting view: HDPE: cheap tooling at volume, no corrosion, recyclable.

2

Intake: conveyor or suction for PoP slurry and scum?

Starting view: Conveyor for solids; test a suction head for algae scum.

3

Perception: radar + cameras, or add LiDAR?

Starting view: Radar + stereo cameras; add LiDAR only if glare tests fail.

4

Charging: auto-dock contacts or manual pack swap?

Starting view: Manual swap in the pilot; auto-dock in Phase 2.

5

Security: how do we prevent theft and vandalism?

Starting view: GPS tracking, tamper alarms, lockable bays, night mooring at docks.

6

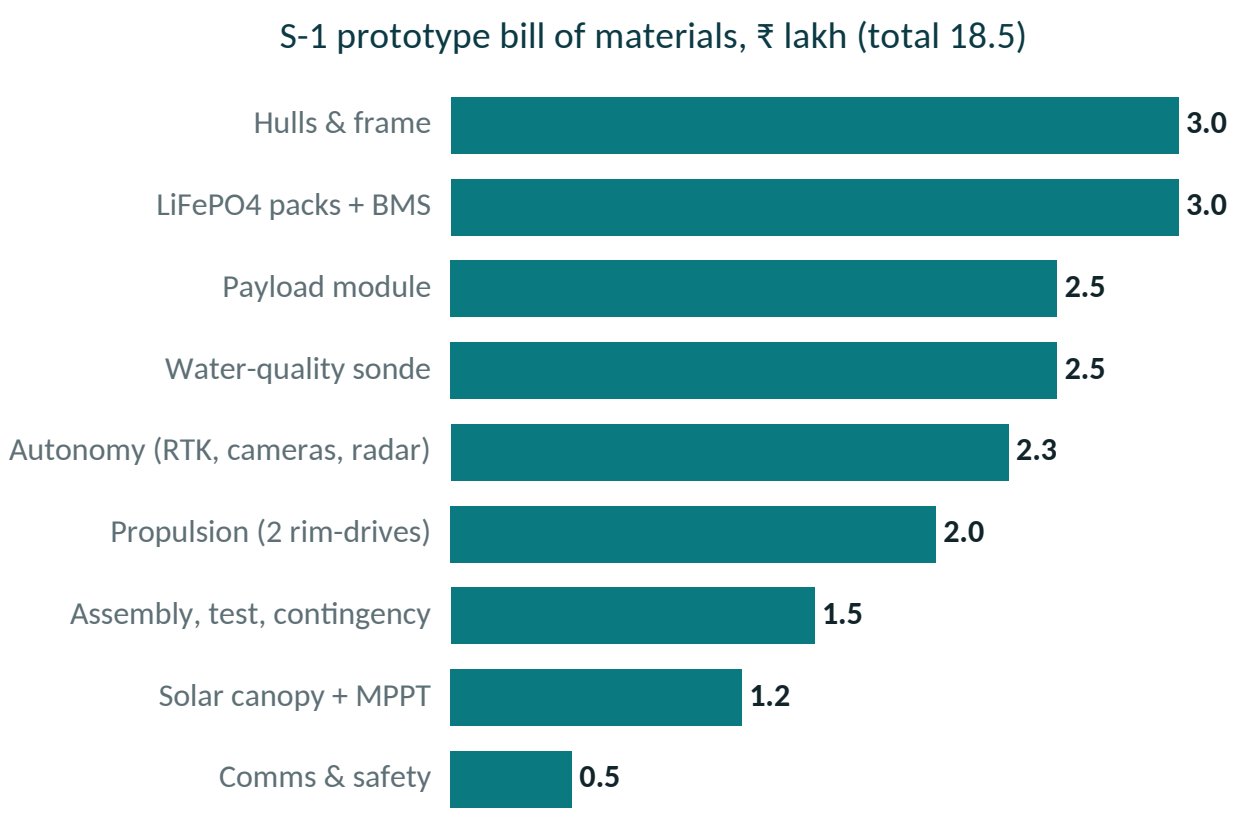
Product line: one hull or two dedicated SKUs?

Starting view: One hull, two swappable payloads, 15-minute swap.

DELIVERY · COST

What it costs

The boats are cheap enough to pilot now; the plant needs public financing (DBO or Hybrid Annuity).



₹18.5 lakh

S-1 prototype, falling to ≤ ₹12 lakh at 50+ units

~₹0.75 crore

Phase 1 pilot: 3 boats + test dock

~₹2.4 crore

Phase 2 fleet: 12 boats + 2 solar docks

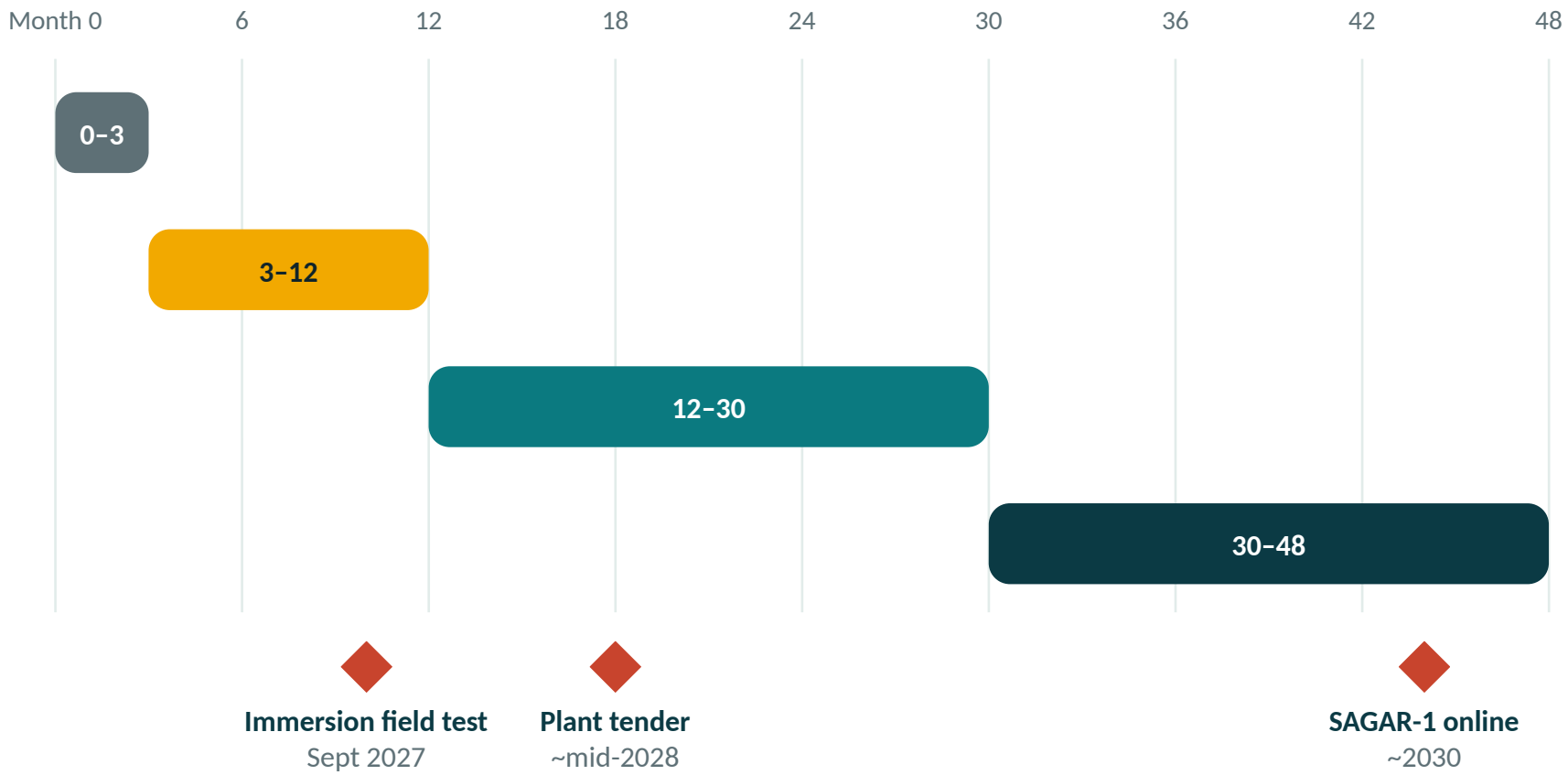
₹450–550 crore

SAGAR-1 plant (DBO / Hybrid Annuity)

Indicative 2026 India pricing for discussion, not quotations. Plant benchmark: ≈ ₹3.4 cr/MLD (HMWSSB 17-STP package) plus an MBR and 4th-stage premium.

48 months from survey to clean inflow

The boats prove the concept at the 2027 immersion season while the plant is designed and built.



DELIVERY · TARGETS

How we'll know it worked

Every target is tied to a CPCB criterion and measured live on the public dashboard.

Untreated dry-weather inflow

0

Today: **≥ 40 MLD**

SAGAR-1 + interceptors on all 4 nalas

Dissolved oxygen

≥ 4 mg/L

Today: **~1.9 mg/L**

CPCB Class D: fish survive

BOD

≤ 10 mg/L

Today: **24–76 mg/L**

Then ≤ 3 mg/L (bathing) long-term

Total coliform

< 500 MPN

Today: **1,600 MPN**

CPCB bathing criterion, per 100 mL

Floating debris

−90%

Today: **not measured**

Camera survey index, set in Phase 0

Public data

Live 24×7

Today: **none**

Open dashboard of every sonde and buoy

Next steps

Design

Freeze the S-1/O-1 platform architecture and settle the six open questions

Weeks 0–4

Build

2 × S-1 + 1 × O-1 prototypes; tank test, then a lake pilot at Lumbini Park

Months 3–9

Partner

MoU with HMDA (lake), TGPCB (monitoring) and HMWSSB (sewerage) for the pilot and the Phase 0 survey

Now

Goal: solar boats on Hussain Sagar for the Ganesh immersion, September 2027.

APPENDIX

Sources

Full links in research/hussain-sagar-pollution-data.md. TGPCB figures are as reported by the press.

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